DATA IS NOT NEUTRAL:

Toward a Political Economy of Business Intelligence

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ABSTRACT

Business Intelligence (BI) systems are conventionally framed as neutral technical instruments that translate organizational data into actionable managerial knowledge. This paper challenges that assumption by applying a political economy perspective to the design, deployment, and governance of BI tools. Drawing on surveillance capitalism theory (Zuboff, 2019), platform capitalism scholarship (Srnicek, 2017), and critical data studies (boyd & Crawford, 2012; Kitchin, 2014), the article argues that BI systems are inherently political artifacts: they encode power relations, reproduce structural inequalities, and serve particular economic interests while rendering others invisible. To operationalize this argument, the paper introduces and extends the BI Power-Alignment Framework (BPAF), a three-layered analytical model examining the Extraction, Representation, and Execution dimensions of organizational data practice. The framework is applied to two contemporary BI domains—algorithmic workforce management and AI-generated executive insights—to demonstrate how seemingly neutral dashboards and KPIs function as instruments of capital accumulation and managerial control. The analysis reveals structural asymmetries in data access, metric design authority, and the distribution of algorithmic benefits across organizational hierarchies. The paper concludes by articulating a research agenda for emancipatory BI, proposing six empirical and theoretical research questions and offering implications for BI practitioners, organizational ethicists, and data governance scholars.

Keywords: business intelligence; political economy; surveillance capitalism; platform capitalism; data ethics; algorithmic governance; KPI design; power dynamics; data governance; emancipatory analytics

1. INTRODUCTION

Open any modern business intelligence dashboard, and you encounter an apparently frictionless epistemic universe. Bar charts ascend optimistically, gauges glow in reassuring greens, and summary tables reduce millions of human transactions to a handful of key performance indicators (KPIs). The implicit communicative contract of the dashboard is one of objectivity: that the organization's performance has been faithfully recorded, neutrally processed, and transparently presented.

This assumption of neutrality is pervasive, institutionally reinforced, and analytically dangerous. The premise of this paper is straightforward but consequential: Business Intelligence is not a mirror reflecting an independent organizational reality. It is a sociotechnical construction—an active, interested, and politically embedded process that creates, selects, and communicates particular versions of organizational truth while suppressing others.

This argument is not merely epistemological. It has material implications. The choice of which metrics to track, which populations to surveil, which algorithmic recommendations to automate, and which stakeholders gain interpretive authority over data—these are not technical decisions. They are political ones, with distributive consequences for labor, capital, and organizational power. As Bowker and Star (1999, p. 5) observed in their seminal work on classification systems, 'to classify is to value.' Every BI architecture is, in this sense, a value system operationalized in code.

The urgency of this critique is amplified by three converging developments in contemporary data practice. First, BI has expanded from static reporting into real-time, AI-augmented decision-support, vastly increasing the speed and scope of organizational surveillance. Second, the infrastructure of BI has consolidated around a small number of cloud platforms and proprietary analytics ecosystems—what Srnicek (2017) terms 'platform capitalism'—creating new forms of vendor dependency and data enclosure. Third, the rhetoric of algorithmic objectivity has become a powerful ideological tool, deflecting organizational accountability by attributing structural decisions to the neutral arbitration of data.

The paper proceeds as follows. Section 2 reviews the relevant literature spanning critical data studies, platform capitalism theory, and the political economy of technology. Section 3 introduces and formally elaborates the BI Power-Alignment Framework (BPAF), a three-tiered analytical model for evaluating the political dimensions of BI systems. Section 4 applies this framework to key domains of contemporary BI practice. Section 5 discusses implications, limitations, and a proposed research agenda. Section 6 concludes with contributions to BI scholarship and practice.

2. LITERATURE REVIEW

2.1 The Myth of Data Neutrality in Critical Data Studies

The notion that data constitutes an objective, pre-theoretical record of reality has been systematically dismantled across multiple scholarly traditions. In science and technology studies (STS), Haraway's (1988) critique of the 'god trick'—the view from nowhere that presents situated knowledge as universal—established that all knowledge-production, including quantitative measurement, reflects the social position and interests of its producers. This insight was extended to organizational data systems by Bowker and Star (1999), who demonstrated that information infrastructures encode the classificatory priorities of their designers, with profound consequences for those who are counted, categorized, or omitted.

Within the emerging field of critical data studies, boyd and Crawford (2012) advanced a foundational critique of 'big data' epistemology, arguing that large datasets do not speak for themselves but are always interpreted through frameworks that carry normative and political assumptions. Kitchin (2014) elaborated this position by distinguishing between 'raw' and 'cooked' data, arguing that all data are 'capta'—taken actively, according to specific protocols, from a world that is always richer than any recording system can capture. This ontological argument has direct implications for BI: what is measured in a dashboard is always a selection, and the act of selection is a political act.

O'Neil (2016) brought this critique to bear on algorithmic systems in organizational settings, documenting how models built on historical data systematically perpetuate and amplify the structural inequalities embedded in that data. Noble (2018) extended this analysis to search and recommendation algorithms, revealing how technically neutral systems reproduce racial and gender hierarchies. These works collectively establish that data systems—however sophisticated their mathematics—are not transcendent of the social relations in which they are embedded.

In the BI literature specifically, critiques of metric design have been advanced by practitioners such as Pfeffer and Sutton (2006), who warned of the dangers of treating quantified proxies as equivalent to the complex organizational realities they are designed to represent. The Goodhart's Law problematic—'when a measure becomes a target, it ceases to be a good measure' (Strathern, 1997)—has been extensively documented in organizational settings from healthcare to financial services, revealing how the political authority vested in KPIs generates systematic distortions in organizational behavior.

2.2 Surveillance Capitalism and Platform Dependency

Zuboff's (2019) theory of surveillance capitalism provides the most comprehensive account of the political economy of data extraction in contemporary capitalism. Zuboff argues that the dominant tech platforms have developed a new economic logic in which human behavioral data is extracted at scale, processed into 'prediction products,' and sold to advertisers and institutional clients seeking to influence future behavior. The critical contribution of this framework for BI analysis lies in its account of the asymmetric power relations embedded in data systems: those who generate data—employees, customers, citizens—are rarely its primary beneficiaries.

This logic of behavioral surplus extraction has migrated from consumer platforms into the organizational interior. Andrejevic (2007) identified the emergence of workplace surveillance technologies as an extension of the panoptic logics first theorized by Foucault (1977) in his analysis of institutional discipline. Where Foucault's panopticon operated through the constant possibility of being observed, contemporary AI-enhanced workforce analytics make continuous, granular surveillance the operative reality rather than the disciplinary threat. Platform-based workforce management systems—deployed by firms such as Amazon and Uber—ingest data on keystroke rhythms, communication patterns, delivery velocities, and even physiological indicators, processing them through algorithmic models that trigger managerial interventions with minimal human review (Rosenblat, 2018).

Srnicek's (2017) framework of platform capitalism illuminates the structural dimension of this dynamic. Srnicek identifies 'lean platforms' and 'cloud platforms' as two critical infrastructure types that now mediate the data relationships of most organizations. Modern BI stacks are heavily dependent on cloud platforms—Amazon Web Services, Microsoft Azure, Google Cloud—that function as data landlords, extracting rent from organizations' need to store, process, and access their own operational memory. This creates what Pasquale (2015) terms 'black box' dependencies: organizations become analytically legible to platform providers while the underlying algorithmic logics of those platforms remain opaque.

Couldry and Mejias (2019) extend this analysis into a global framework, arguing that data colonialism—the systematic extraction of human behavioral data from populations in the Global South by platforms headquartered in the Global North—represents a new mode of expropriation structurally analogous to historical colonialism. This framework has particular resonance for BI practitioners and organizations operating in emerging market contexts, where data infrastructure is predominantly imported from external providers while the value generated from local behavioral data flows outward.

2.3 Political Economy, Labor, and the Dashboard

The political economy tradition offers a structural lens for understanding BI as an instrument of capital-labor relations. The Marxist concept of the labor process—developed by Braverman (1974) in his account of the deskilling of industrial work—provides a foundational framework: BI systems, like earlier forms of scientific management, represent the formal subsumption of labor knowledge under capital control. The dashboard that tracks driver idle time or call-center handling duration is not merely measuring performance; it is transferring the knowledge of how work is performed from the worker to the managerial apparatus, enabling closer supervision and tighter control (Weil, 2014).

This dynamic has been analyzed specifically in the context of algorithmic management by Kellogg, Valentine, and Christin (2020), who distinguish between algorithmic systems that monitor, nudge, replace, and discipline workers—identifying a spectrum of intensification that varies by industry, labor market position, and organizational context. Their research demonstrates that the political consequences of BI are not uniform but are mediated by the structural vulnerabilities of different worker populations.

In the Global South context, the political economy of BI carries additional dimensions. Mahdi Amel's (1985) framework of the colonial mode of production—subsequently developed in the tradition of Arabic Marxist scholarship—provides analytical tools for understanding how organizational data systems introduced through multinational corporations and multilateral development institutions can reproduce structural dependencies rather than building indigenous analytical capacity. This framework suggests that the political economy of BI cannot be disaggregated from the political economy of the technological systems through which it is deployed.

Harvey's (2005) concept of 'accumulation by dispossession' has been applied to data by Sadowski (2019), who argues that data extraction from workers, communities, and ecosystems represents a form of primitive accumulation—the conversion of social goods into private capital. In organizational settings, this process is operationalized through BI systems that capture the tacit knowledge of frontline workers and convert it into proprietary algorithmic assets owned by the organization, without corresponding compensation or recognition.

3. THEORETICAL FRAMEWORK: THE EXTENDED BI POWER-ALIGNMENT FRAMEWORK

Building on the theoretical traditions reviewed above, this paper introduces the BI Power-Alignment Framework (BPAF)—a three-tiered analytical model for examining the political economy of organizational data systems. The framework is designed to complement rather than replace existing technical and managerial approaches to BI evaluation; its purpose is to make visible the power relations that conventional BI assessment frameworks render invisible.

The BPAF is grounded in three analytical claims. First, that every BI system operates simultaneously as a technical system and as a political system, with distributional consequences for organizational actors. Second, that these distributional consequences are structured across three analytically separable dimensions: data extraction, knowledge representation, and algorithmic execution. Third, that critical evaluation of a BI system requires systematic interrogation of each dimension in terms of who bears costs, who holds authority, and who accrues benefits.

The framework draws on Foucault's (1980) concept of power/knowledge—the argument that systems of knowledge production are simultaneously systems of power, constituting what can be thought and said within an organizational order—as well as Lukes' (2005) three-dimensional model of power, which extends analysis beyond observable conflicts to include agenda-setting and the shaping of preferences. It is also informed by Lefebvre's (1991) spatial theory, which has been adapted by critical data scholars to analyze how data architectures produce and reproduce social space.

The three layers of the BPAF are formalized as follows:

LAYER 1: THE EXTRACTION LAYER — Who Bears the Burden of Data Production?
Technical Question: How do we collect data efficiently, reliably, and at scale?
Political Economy Question: Who bears the burden—material, temporal, cognitive, and dignity-related—of data production? Is data collected cooperatively (with informed consent and reciprocal benefit) or coercively (under conditions of employment dependency, asymmetric information, or implicit compulsion)?
Key Analytical Dimensions: Consent architecture; surveillance intensity; data subject agency; cross-sectional equity in monitoring burden
Structural Risk: Disproportionate surveillance of frontline and lower-status workers relative to management and executive populations

LAYER 2: THE REPRESENTATION LAYER — Who Defines Organizational Reality?
Technical Question: What is the optimal method for visualizing KPIs and communicating analytical insights?
Political Economy Question: Which stakeholders hold authority over metric design—the selection of what to measure, what to aggregate, and what to exclude from the semantic layer?

Whose definition of organizational success is operationalized, and whose structural reality is rendered invisible?

Key Analytical Dimensions: Stakeholder inclusion in metric design; visibility of negative externalities; treatment of labor conditions, environmental costs, and community impact in the BI semantic layer

Structural Risk: Systematic underrepresentation of costs borne by non-capital stakeholders; reinforcement of short-term financial metrics over systemic organizational health

LAYER 3: THE EXECUTION LAYER — Who Benefits from Algorithmic Action?

Technical Question: How should analytical insights be operationalized into automated decisions and workflow interventions?

Political Economy Question: When an AI system generates an insight or triggers an automated decision, who absorbs the cost and who accrues the margin? Does algorithmic execution consolidate organizational power and distribute the productivity of analytical intelligence equitably, or does it intensify structural asymmetries?

Key Analytical Dimensions: Distribution of algorithmic benefits across organizational hierarchy; human review thresholds; contestability mechanisms for affected workers; accountability structures for automated decisions

Structural Risk: Automation of structural inequality under conditions of algorithmic neutrality rhetoric; erosion of human judgment in high-stakes labor and resource allocation decisions

The three layers are analytically sequential but empirically interdependent: the extraction layer determines what data enters the system; the representation layer determines how that data constitutes organizational knowledge; and the execution layer determines how that knowledge is converted into power. Critically, ideological work occurs at all three layers—in the naturalization of monitoring regimes, the reification of particular metrics as objective truth, and the delegation of structural decisions to algorithmic systems.

The BPAF is designed to function as both a diagnostic and an evaluative instrument. As a diagnostic, it enables analysts, practitioners, and governance stakeholders to systematically identify where power asymmetries are encoded in a given BI system. As an evaluative instrument, it provides criteria for assessing whether a proposed BI architecture represents an equitable distribution of the costs and benefits of organizational data practice.

4. ANALYSIS: APPLYING THE BPAF TO CONTEMPORARY BI PRACTICE

4.1 KPIs as Instruments of Capital: The Logistics Example

Consider the design of a performance management dashboard for a logistics and delivery network—a domain that has been extensively transformed by AI-driven analytics over the past decade. A canonical BI implementation in this context would track variables including average delivery time, route efficiency, fuel consumption, idle time, and on-time delivery rate. Each of these metrics appears, on its surface, to be an objective and operationally motivated measure.

Analyzed through the BPAF's Extraction Layer, however, the political economy of this surveillance architecture becomes legible. The system monitors a specific population—frontline delivery workers—with an intensity that is not applied to the managerial and executive layers of the same organization. The asymmetry of surveillance is structural: workers generate data as a condition of employment; managers consume it as an instrument of control. As Kellogg, Valentine, and Christin (2020, p. 371) document, 'algorithmic management systems monitor, evaluate, and direct workers by collecting data on their behavior, processing it through algorithms, and using the results to control their performance'—a dynamic that reproduces, in digital form, the labor process hierarchies analyzed by Braverman (1974).

At the Representation Layer, the standard logistics dashboard encodes a specific theory of value: that driver performance is adequately captured by velocity and efficiency metrics, and that factors such as road safety precautions, driver fatigue, local traffic conditions, and worker well-being are system noise rather than system signal. This representational choice is not politically innocent. By designing the semantic layer around capital efficiency—reducing delivery time, maximizing route density, minimizing idle time—the dashboard constructs a version of organizational reality in which labor friction is a problem to be optimized rather than a legitimate constraint to be accommodated. The metrics that would reveal the costs of this optimization—accident rates correlated with algorithmic pressure, worker burnout indices, longitudinal health outcomes—are systematically absent from the standard BI architecture.

At the Execution Layer, the consequences of this representational architecture are operationalized through algorithmic decision-making. In the most advanced implementations—such as those deployed by Amazon Delivery Service Partners (DSP) and gig economy platforms such as Uber and Lyft—algorithmic systems autonomously trigger disciplinary interventions, contract adjustments, and terminations based on performance metrics (Rosenblat, 2018). The automation of these consequential decisions eliminates the human review processes that might incorporate contextual judgment, worker advocacy, or proportionality considerations. Workers subjected to these systems face what Pasquale (2015) describes as 'black box' governance: opaque algorithms that determine their livelihoods without explanation, transparency, or contestability.

This analysis illustrates the central claim of the paper: the logistics dashboard does not neutrally reflect organizational reality. It constructs a particular version of organizational reality—one organized around the velocity of capital accumulation—and deploys that construction to govern the labor force in ways that serve managerial and shareholder interests while externalizing the costs of operational efficiency onto workers and, through secondary effects, onto surrounding communities and environments.

4.2 AI-Generated Insights and the Automation of Ideology

The integration of AI-generated insights into contemporary BI platforms—through features such as Microsoft Power BI's 'Quick Insights,' Tableau's 'Explain Data,' and Qlik's 'Insight Advisor'—represents a qualitative escalation of the political dynamics analyzed above. These systems do not merely present data; they generate and present interpretations of data, framing organizational situations and recommending courses of action through algorithmic processes that are largely opaque to their organizational users.

The political economy of AI-generated insights operates most acutely at the Representation Layer of the BPAF. When a BI platform's AI module identifies a 'workforce optimization opportunity' and recommends headcount reduction in a particular operational unit, it is not performing a politically neutral calculation. It is operationalizing a specific economic ideology—one that treats labor as a cost variable rather than an organizational asset—through the rhetorical authority of algorithmic objectivity. As O'Neil (2016, p. 3) argues, 'algorithms are opinions embedded in mathematics.' The AI's recommendation inherits all the assumptions, priorities, and structural biases encoded in the training data and optimization functions on which it was built.

The ideological function of AI insights is amplified by what Eubanks (2018) terms the 'automation bias'—the documented tendency of human decision-makers to defer to automated recommendations, particularly in organizational contexts where the social authority of data and algorithmic systems is high. In the context of BI, automation bias means that the political choices embedded in AI-generated insights tend to escape critical scrutiny precisely because they are presented as the output of a neutral analytical process rather than as the recommendations of a politically interested human actor.

There is also a structural asymmetry in the distribution of AI-generated insight. The AI features embedded in contemporary BI platforms are predominantly designed to generate insights for executive and managerial users—to support strategic decision-making, cost optimization, and market positioning. They are not, in general, designed to generate insights for workers about the conditions of their own labor, the distribution of organizational surplus, or the environmental impact of operational decisions. This asymmetry in the design of AI insight systems reflects and reinforces the power hierarchy that characterizes the organizations these systems serve.

4.3 Data Governance, Structural Asymmetry, and the Emerging Market Context

The political economy of BI assumes particular dimensions in emerging market organizational contexts, where the asymmetries analyzed above are compounded by structural inequalities in global data infrastructure. Organizations in sub-Saharan Africa, the MENA region, and South and Southeast Asia are predominantly dependent on BI infrastructure—cloud platforms, analytics tools, algorithmic models—developed and controlled by firms headquartered in the United States and Europe.

This structural dependency has several analytically significant consequences. First, at the Extraction Layer, data generated by organizations and workers in emerging markets is processed and stored on infrastructure owned by foreign platform corporations, creating a form of data sovereignty deficit—what Couldry and Mejias (2019) describe as the 'asymmetric logic of data colonialism.' The commercial behavioral data of consumers, workers, and organizations in Sudan, Nigeria, or Bangladesh flows into the training corpora and recommendation systems of platforms in which those populations have no ownership stake or governance voice.

Second, at the Representation Layer, the KPI templates, industry benchmarks, and algorithmic models that constitute the default semantic layer of global BI platforms encode assumptions calibrated for the economic and institutional contexts of developed market firms. Performance benchmarks derived from US or European operational contexts may be structurally inappropriate for application in emerging market environments with different labor market conditions, infrastructure constraints, and institutional frameworks. The application of these benchmarks through BI systems can generate systematic misrepresentations of organizational performance.

Third, at the Execution Layer, the algorithmic recommendations generated by global BI platforms for organizations in emerging markets may reflect optimization logics that serve the interests of multinational capital rather than local organizational or community development. In sectors such as microfinance, agricultural supply chains, and health systems—where BI adoption is accelerating across the Global South—these dynamics have implications that extend well beyond organizational management to community welfare and economic development.

5. DISCUSSION

5.1 Toward Emancipatory Business Intelligence

The analysis presented above is not intended to generate a counsel of despair regarding BI practice. Rather, it is intended to establish the conditions of possibility for a more ethically rigorous and structurally aware approach to organizational data work—what we term 'emancipatory BI.' This concept draws on the tradition of emancipatory social science (Habermas, 1972) and its application in management scholarship (Alvesson & Willmott, 1992), which argues that critical analysis is not merely descriptive but serves an emancipatory function: making visible the structural constraints that appear as natural givens, and thereby opening space for their transformation.

Four practical implications follow from the BPAF analysis for BI practitioners and organizational leaders. First, the democratization of metric design: the authority over KPI selection and semantic layer architecture should not be monopolized by executive and managerial stakeholders. Participatory metric design processes—which systematically include the workers, communities, and other stakeholders whose activities are being measured—are more likely to produce representational frameworks that capture the full complexity of organizational performance, including its distributional dimensions.

Second, the systematic auditing of AI-generated insights for distributional consequences: before automating an algorithmic recommendation, practitioners and governance stakeholders should trace the distributional logic of that recommendation—identifying who bears the cost and who accrues the margin, and whether those distributions are consistent with the organization's stated values and legal obligations. This is not merely an ethical imperative; it is increasingly a legal one, as data protection regulations in multiple jurisdictions establish rights to explanation and contestability for algorithmic decisions that affect individuals.

Third, the cultivation of what we term 'productive data friction': the intentional introduction of human review and deliberation at the Execution Layer for decisions with significant consequences for labor, resource allocation, and community welfare. The rhetoric of real-time, automated execution that dominates contemporary BI discourse needs to be interrogated for the distributional implications of removing human judgment from consequential organizational decisions.

Fourth, the construction of counter-dashboards that track organizational dimensions currently excluded from the standard BI semantic layer: the distribution of compensation and organizational surplus, the environmental costs of operational decisions, the correlation between output intensification and worker health outcomes, and the long-term community impacts of organizational resource extraction.

5.2 Research Agenda: Questions and Hypotheses for Empirical and Theoretical Study

The BPAF and the analysis presented above generate a number of research questions that warrant systematic empirical and theoretical investigation. The following questions are offered as contributions to a developing research agenda in critical BI studies:

RQ1: Structural Asymmetry in Surveillance

To what extent is the intensity and granularity of BI-enabled monitoring asymmetrically distributed across organizational hierarchy, and what organizational, sectoral, and institutional factors moderate this asymmetry? Hypothesis: Organizations operating in highly competitive product markets, or in sectors with high labor cost shares, will exhibit greater surveillance asymmetry (more intensive monitoring of frontline workers relative to management) than organizations in innovation-intensive or knowledge-intensive sectors.

RQ2: Participatory Metric Design and Organizational Performance

Does participatory metric design—the systematic inclusion of non-managerial stakeholders in KPI selection and semantic layer architecture—produce measurably different organizational outcomes in terms of employee engagement, operational sustainability, and stakeholder trust, compared to conventional top-down metric design? This question invites longitudinal quasi-experimental research designs comparing organizations that have adopted participatory metric design with matched controls.

RQ3: Automation Bias in AI-Generated BI Insights

How does the framing of algorithmic recommendations in contemporary BI platforms (as 'AI insights' versus as 'analyst suggestions') affect the degree of critical scrutiny applied by organizational decision-makers, and what individual and organizational factors moderate this effect? This question can be addressed through experimental research designs using vignette scenarios with organizational decision-maker samples.

RQ4: Data Sovereignty and Emerging Market BI Governance

How do organizations in emerging market contexts negotiate the tension between the operational benefits of global BI platform adoption and the data sovereignty, interpretive autonomy, and governance risks that accompany structural dependency on foreign platform providers? This question invites qualitative comparative case study research across multiple regional and sectoral contexts.

RQ5: The BPAF as Organizational Audit Instrument

Can the BI Power-Alignment Framework be operationalized as a reliable and valid organizational audit instrument, and does its application generate actionable insights for BI governance reform? This question invites instrument development research, including the construction and validation of BPAF assessment scales for use in organizational self-assessment and third-party audit contexts.

RQ6: Distributional Outcomes of Counter-Dashboard Implementation

When organizations implement 'counter-dashboards'—BI systems specifically designed to track distributional outcomes, externalities, and stakeholder impacts currently excluded from conventional performance reporting—what effects are observed on managerial decision-making, organizational resource allocation, and reported stakeholder outcomes? This question invites both case study research and, over longer time horizons, longitudinal panel designs.

5.3 Limitations

This paper has several notable limitations. The BPAF, as presented here, is a conceptual rather than empirically validated framework. Its analytical utility is demonstrated through illustrative application to specific BI domains, but its reliability, validity, and generalizability as an organizational assessment tool require systematic empirical testing. The illustrative cases selected—logistics workforce management and AI-generated executive insights—are drawn primarily from the documented experience of large, predominantly US-based technology platforms. The applicability of the analysis to small and medium-sized enterprises, to public sector organizations, and to a wider range of regional and cultural contexts requires further investigation. Finally, the paper's focus on the political economy of BI necessarily de-emphasizes the significant organizational benefits that well-designed BI systems can deliver. A comprehensive assessment of BI would need to account for both the emancipatory and the constraining potentials of data-driven organizational management.

6. CONCLUSION

This paper has argued that Business Intelligence systems are not neutral technical instruments but politically embedded sociotechnical constructions that encode and reproduce specific distributions of organizational power, authority, and value. Drawing on critical data studies, surveillance capitalism theory, platform capitalism scholarship, and the political economy tradition, we have demonstrated that the metrics organizations track, the platforms they deploy, and the algorithms they automate are deeply implicated in the reproduction of capital-labor relations, the governance of organizational knowledge, and the distribution of the benefits and costs of data-driven operations.

The BI Power-Alignment Framework offers a systematic analytical instrument for making these power relations visible across three dimensions of organizational data practice: the extraction of data from human activity, the representation of that data as organizational knowledge, and the execution of automated decisions on the basis of that knowledge. Applied to the domains of logistics workforce management, AI-generated BI insights, and emerging market data governance, the BPAF reveals structural asymmetries that are obscured by the dominant technocratic discourse of data neutrality.

The practical implications of this analysis are significant. BI practitioners who accept uncritically the metrics, platforms, and algorithmic recommendations offered by their tools risk becoming unwitting participants in the reproduction of structural inequality within and beyond their organizations. A more critical, ethically rigorous approach to BI—one that democratizes metric design, audits AI recommendations for distributional consequences, introduces productive friction at the execution layer, and expands the BI semantic layer to include currently invisible costs and externalities—is both analytically and normatively justified.

For BI scholarship, the central contribution of this paper is to establish political economy as a legitimate and necessary analytical framework for the study of organizational data systems. The question of whether BI serves justice as well as efficiency, equity as well as accuracy, and democratic accountability as well as managerial control, is not peripheral to the field. It is constitutive of its social purpose.

The data analysts, BI architects, and organizational leaders who design the dashboards of the contemporary firm are not merely processing information. They are building the epistemic infrastructure through which organizational reality is constituted, contested, and governed. That is a political responsibility as much as a technical one.

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